Numerical implications in audio watermarking

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Goals: **(i) imperceptibility**, (ii) payload, (iii) security, **(iv) robustness**

Imperceptibility: using perceptual evaluation of audio quality (PEAQ) measurement algorithm specified in ITU-R BS.1387,

signal-to-noise ratio $SNR = 10\log_{10}\frac{\sum_{n=1}^{N} Y^2(n)}{\sum_{n=1}^{N}[Y(n)-Y'(n)]^2}$

Y(n) and Y'(n) original and watermarked audio signal, respectively

SG	ODG	Description of impairments	Quality
5.0	0.0	Imperceptible	Excellent
4.0	−1.0	Perceptible, but not annoying	Good
3.0	−2.0	Slightly annoying	Fair
2.0	−3.0	Annoying	Poor
1.0	−4.0	Very annoying	Bad

Robustness: normalized correlation, bit error rate

$$NC(W,\tilde{W}) = \frac{\sum_{i=1}^{M}\sum_{j=1}^{M} W(i,j)\tilde{W}(i,j)}{\sqrt{\sum_{i=1}^{M}\sum_{j=1}^{M} W^2(i,j)}\sqrt{\sum_{i=1}^{M}\sum_{j=1}^{M} \tilde{W}^2(i,j)}},\quad BER(W,\tilde{W}) = \frac{\text{Number of error bits}}{\text{Number of total bits}} = \frac{\sum_{i=1}^{M}\sum_{j=1}^{M} W(i,j)\oplus\tilde{W}(i,j)}{M\times M}$$

W and $\tilde{W}$ original and the extracted M x M binary watermarks, respectively

Results: numerical conditions on W such that audio watermarking algorithms preserve imperceptibility and robustness; less costly algorithms that reasonably achieve the same goals

Algorithms and attack methods

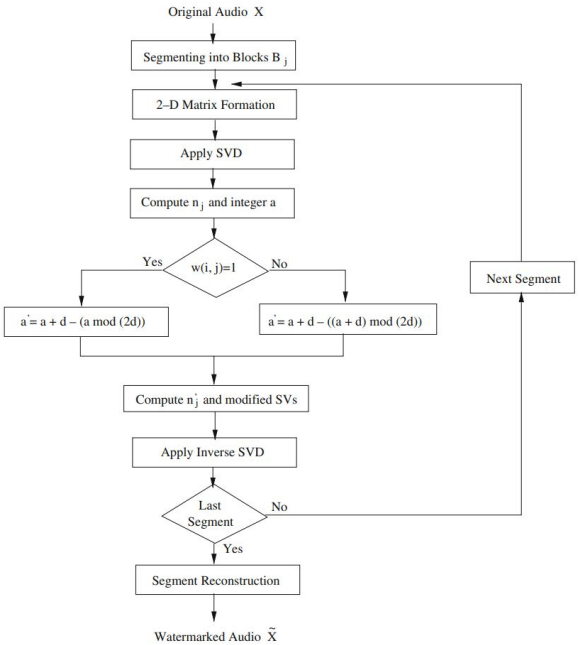


Fig. 1 Watermark embedding

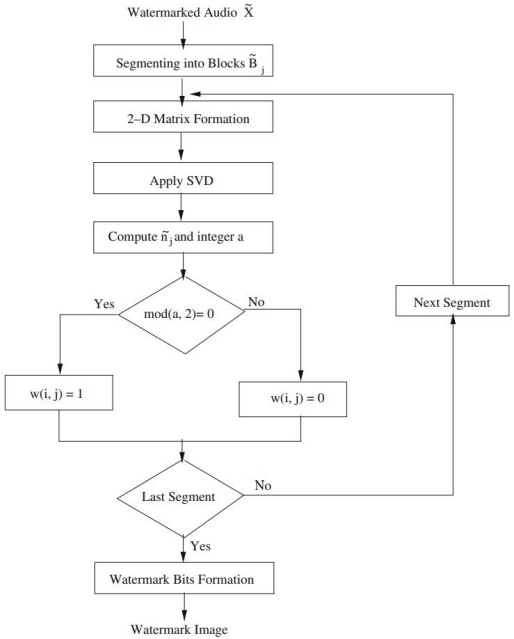


Fig. 2 Watermark extraction

(A) Additive white Gaussian noise

(B) Resampling

(C) Low-pass filtering

(D) Requantization

(E) MP3 compression 64 kbps

(F) MP3 compression 32 kbps

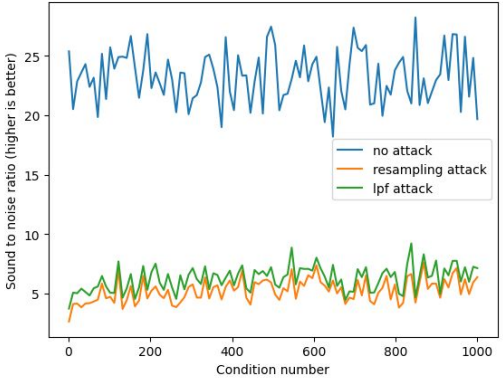
(G) Cropping

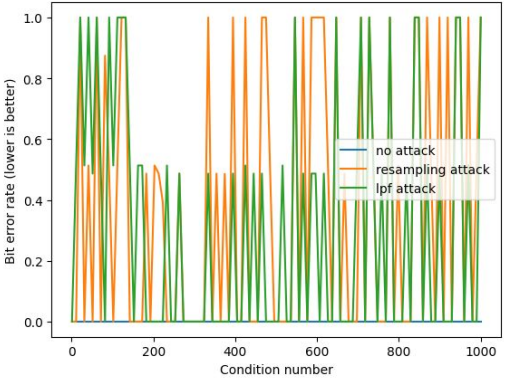
(H) Echo addition

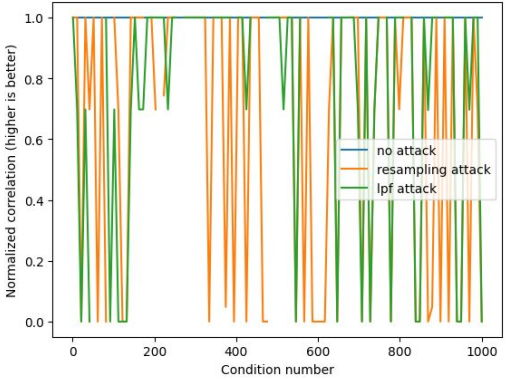
(I) Denoising

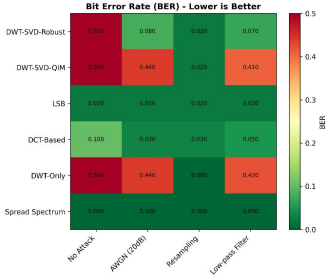
potential components: Discrete Wavelet Transform, Independent Component Analysis, QR decomposition

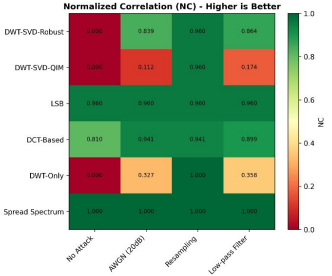
Results

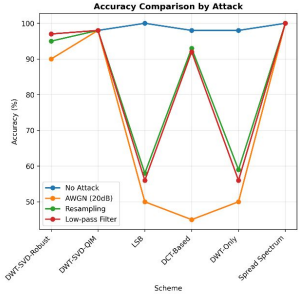

Sound to noise ratio (higher is better)
Condition number
no attack
resampling attack
lpf attack

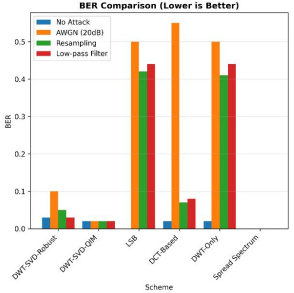

Bit error rate (lower is better)
Condition number
no attack
resampling attack
lpf attack

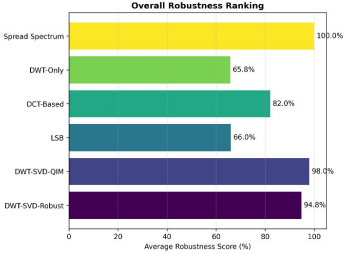

Normalized correlation (higher is better)
Condition number
no attack
resampling attack
lpf attack

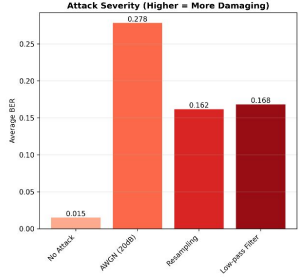

Bit Error Rate (BER) - Lower is Better
DWT-SVD-Robust
DWT-SVD-QIM
LSB
DCT-Based
DWT-Only
Spread Spectrum
No Attack
AWGN (20dB)
Resampling
Low-pass Filter
BER

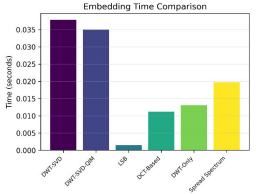

Normalized Correlation (NC) - Higher is Better
DWT-SVD-Robust
DWT-SVD-QIM
LSB
DCT-Based
DWT-Only
Spread Spectrum
No Attack
AWGN (20dB)
Resampling
Low-pass Filter
NC

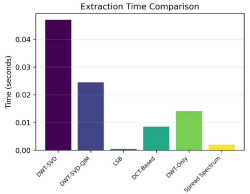

Accuracy Comparison by Attack
Accuracy (%)
Scheme
No Attack
AWGN (20dB)
Resampling
Low-pass Filter
DWT-SVD-Robust
DWT-SVD-QIM
LSB
DCT-Based
DWT-Only
Spread Spectrum

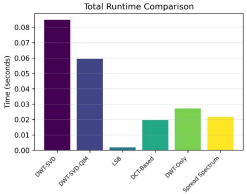

BER Comparison (Lower is Better)
No Attack
AWGN (20dB)
Resampling
Low-pass Filter
BER
Scheme
DWT-SVD-Robust
DWT-SVD-QIM
LSB
DCT-Based
DWT-Only
Spread Spectrum

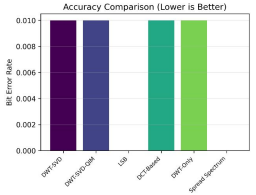

Overall Robustness Ranking
Spread Spectrum
DWT-Only
DCT-Based
LSB
DWT-SVD-QIM
DWT-SVD-Robust
100.0%
65.8%
82.0%
66.0%
98.0%
94.8%
Average Robustness Score (%)

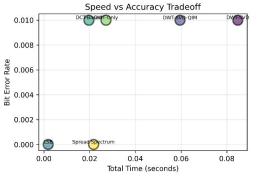

Attack Severity (Higher = More Damaging)
Average BER
0.015
0.278
0.162
0.168
No Attack
AWGN (20dB)
Resampling
Low-pass Filter

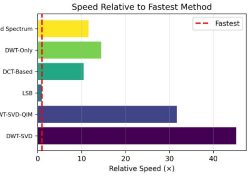

Embedding Time Comparison
Time (seconds)
DWT-SVD
DWT-SVD-QIM
LSB
DCT-Based
DWT-Only
Spread Spectrum

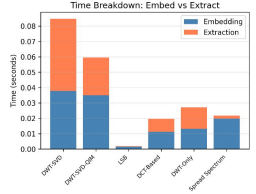

Extraction Time Comparison
Time (seconds)
DWT-SVD
DWT-SVD-QIM
LSB
DCT-Based
DWT-Only
Spread Spectrum

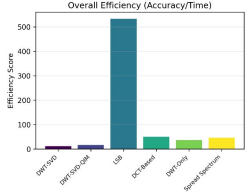

Total Runtime Comparison
Time (seconds)
DWT-SVD
DWT-SVD-QIM
LSB
DCT-Based
DWT-Only
Spread Spectrum

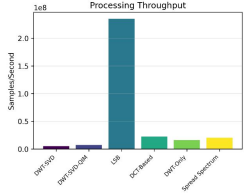

Accuracy Comparison (Lower is Better)
Bit Error Rate
DWT-SVD
DWT-SVD-QIM
LSB
DCT-Based
DWT-Only
Spread Spectrum


Speed vs Accuracy Tradeoff
Bit Error Rate
Total Time (seconds)


Speed Relative to Fastest Method
Fastest
Spread Spectrum
DWT-Only
DCT-Based
LSB
DWT-SVD-QIM
DWT-SVD
Relative Speed (x)


Time Breakdown: Embed vs Extract
Embedding
Extraction
Time (seconds)
DWT-SVD
DWT-SVD-QIM
LSB
DCT-Based
DWT-Only
Spread Spectrum


Overall Efficiency (Accuracy/Time)
Efficiency Score
DWT-SVD
DWT-SVD-QIM
LSB
DCT-Based
DWT-Only
Spread Spectrum


Processing Throughput
1e8
Samples/Second
DWT-SVD
DWT-SVD-QIM
LSB
DCT-Based
DWT-Only
Spread Spectrum